EnergyLens — EXIST-Gründungsstipendium Application Draft

Energy intelligence & decision-support for European commercial buildings, on Microsoft Fabric

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Contents

Note on this draft. This is a working application package, written to be refined with the BSBI Transfer Office and the academic mentor before submission. Figures marked *target* or *illustrative* are pre-revenue assumptions to be validated in pilots, not results. The program is the *EXIST-Gründungsstipendium* (formerly Gründerstipendium); EXIST accepts applications in English.

1. Idea sketch (the 60-second read)

The problem. Buildings consume about **40 % of EU energy** and produce about **36 % of energy-related CO_2** — the single largest sector. Yet most commercial-building portfolios are still managed with monthly utility bills and spreadsheets. They cannot see the anomalies that waste energy nightly, cannot track the wall of tightening EU regulation (EPBD minimum standards, EnEfG, GEG, CSRD/ESRS, the EU Battery Regulation), and cannot prioritise the cheapest decarbonisation measures. The large enterprise vendors (Schneider, Siemens, Measurabl) serve the top of the market; the **mid-market is underserved**.

The solution. EnergyLens is a Microsoft Fabric-native energy-intelligence and decision-support platform for commercial buildings. It unifies meter, sensor, weather, solar and battery data into one auditable data model and surfaces it through three layers: a **9-page analytics report**, a **multi-tenant web application**, and an **AI copilot** that answers questions over the data using *tool use* — running real queries rather than summarising text — and can take action, not just describe it. On top sits a **compliance engine** that scores each building against CRREM decarbonisation pathways, EnEfG, GEG and CSRD/ESRS E1.

The innovation. EnergyLens is not a dashboard. Its defensible core is (a) an **agentic AI copilot** that reasons over a building-energy Lakehouse through tool use, (b) a **regulator-grade, fully traceable carbon-lineage methodology** for CSRD/ESRS reporting, and (c) **multi-protocol IoT fault detection** (ASHRAE Guideline 36) delivered at mid-market prices. The EXIST year develops these from a working prototype into a defensible, validated product (see §4).

The market. The Building Energy Management Systems (BEMS) software market is independently estimated at roughly **US $8.4 bn in 2026, growing to ~$24 bn by 2035 (~8.7 % CAGR)**. Our focus is the **DACH mid-market** — the highest-value EU compliance market and our home market.

Why us. The founder combines genuine **energy-engineering domain knowledge** (B.Sc. Energy Engineering, M.Sc. Energy Management at BSBI) with **modern data-engineering skill** (Microsoft DP-600), and has already built the full platform — pipeline, analytics, web app and AI copilot — solo, end-to-end, in about five weeks. This domain × data-engineering intersection is rare.

The 12-month goal. Convert a working prototype + a BSBI campus pilot into an **incorporated company (UG)** with **5+ paying customers** and **€5–10k MRR**, a published carbon-methodology paper, and a defensible agentic-AI product — i.e. a fundable, post-grant business.

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2. The problem & why now

European commercial real estate sits at the centre of the EU's climate agenda, and the regulatory pressure is now **binding, not aspirational**:

- **EPBD (recast, Directive (EU) 2024/1275):** national minimum-energy-performance standards for non-residential buildings by **1 January 2027**; the worst-performing **16 %** must be improved by **2030** and **26 %** by **2033**.
- **EnEfG (Germany):** organisations above ~250 employees or 7.5 GWh/year must run ISO 50001-grade energy management.
- **GEG §71 (Germany):** new heating systems must use ≥ 65 % renewable energy.
- **CSRD / ESRS E1:** sustainability disclosure (narrowed by the 2026 Omnibus to firms > 1,000 employees & > €450 m turnover, with smaller firms reporting voluntarily via VSME).
- **EU Battery Regulation (2023/1542):** carbon-footprint and recycled-content disclosure, with a digital battery passport from 2027.

At the same time, the structural reasons to act all arrived together: the post-2022 energy-price shock made energy a board-level topic; **Microsoft Fabric** reached general availability (2024); IoT sensor costs collapsed; and AI made pattern recognition affordable. Owners face rising compliance exposure with no modern, affordable tooling — a real, time-bound demand.

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3. The solution & what is already built

A central point for this application: **the product is not a concept — it already runs end-to-end.** That de-risks the venture and is itself evidence of execution capability.

Already built (April–May 2026, solo):

- A full **Microsoft Fabric medallion architecture** (Bronze → Silver → Gold) on OneLake, with ~57 Lakehouse tables and a single sourced reference layer for emission factors and tariffs.
- A **9-page Power BI report** (consumption, building deep-dive, anomalies, forecasting, decision support, sustainability/GHG, HVAC, real-time IoT, battery ROI) on a custom DAX model, reading live via DirectLake.
- A **web application** (Next.js + FastAPI + Azure PostgreSQL) with three-provider authentication, a three-layer access model (row-level security, module gating, subscription tiers), custom pages reading Fabric through the SQL Analytics Endpoint, a five-feature EU-compliance hub, PDF export and an audit log.
- An **AI copilot** with six production tools over the Lakehouse, server-sent-events streaming and a provider abstraction (Anthropic / Azure OpenAI / Mock).
- A **real-time IoT / fault-detection layer** implementing ASHRAE Guideline 36 (12 diagnostic rules, ~31 sensor types, multi-protocol adapters for BACnet/Modbus/MQTT).
- Real external data already integrated (Open-Meteo weather, ENTSO-E prices, ElectricityMaps grid carbon).

The platform currently runs on **ten representative buildings** (Germany, Turkey, Austria, the Netherlands; ~3.5 years; ~693,000 data points). **There are no paying customers yet** — a BSBI campus pilot is in active discussion. The grant is what converts this proven prototype into a validated, incorporated company.

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4. Innovation & unique selling proposition

EXIST funds *innovative, knowledge-based products with a clear unique selling proposition*. EnergyLens's innovation is not a single feature but a combination that no competitor currently offers at the mid-market, plus a research-grade development agenda for the grant year. We are explicit about the distinction between **what already works** and **what the grant develops**, because honesty about maturity is itself a credibility signal to a knowledgeable reviewer.

4.1 The unique combination (already working)

1. **Microsoft Fabric-native, with auditable carbon lineage.** Because every workload reads one copy of the data on OneLake, every disclosed kilogram of CO_2 can be traced back to the raw bill row. This "regulator-grade lineage" is exactly what CSRD/ESRS auditors require and is structurally hard for spreadsheet-based or multi-tool competitors to match.
2. **An AI copilot that uses tool use, not text summaries.** Most BI "copilots" summarise a report page. EnergyLens's copilot runs *real queries* over the Lakehouse through typed tools — and one tool writes back, so it can act, not only describe.
3. **Multi-protocol IoT fault detection at mid-market price.** ASHRAE Guideline 36 automated fault detection (the standard the large BMS vendors use) across BACnet, Modbus and MQTT — delivered as affordable SaaS rather than a six-figure enterprise installation.
4. **A multi-country EU-regulation engine** (DE, AT, NL, TR, EU) that turns findings into ranked, grant-matched actions.

4.2 The innovation agenda (what the EXIST year develops)

The grant year turns the working platform into a defensible, research-grade product along four axes:

- **A — Agentic AI copilot.** Evolve the copilot from answering questions to **proposing and sequencing** retrofit and dispatch plans — an autonomous "decarbonisation advisor" grounded in the building's own data.
- **P — Predictive intelligence.** Move beyond statistical forecasting to **ML-driven predictive maintenance** that detects developing faults before they incur cost.
- **O — Closed-loop optimisation.** Move from *simulation* to *control*: optimal battery dispatch, HVAC setpoint optimisation and automated demand response, aligned with the upcoming EU Demand-Response Network Code.
- **M — Auditable carbon methodology (academic IP).** Formalise the traceable carbon-lineage method into a **publishable methodology**, developed jointly with BSBI — strengthening both the defensibility of the product and the academic-transfer character that EXIST values.

4.3 Defensibility (moat)

The moat is the **compounding combination**: domain-specific data models + an auditable methodology (potential academic IP) + a Fabric-native architecture + an agentic AI layer + an EU-regulatory knowledge base, all aimed at a segment the incumbents under-serve. None of these alone is unassailable; together, and with first-mover reference customers in DACH, they create a defensible position. Microsoft-ecosystem alignment (DP-600, Partner Network track) adds a distribution moat.

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5. Market & business model

5.1 Market size

- **Credible external anchor:** the **BEMS software market** is independently estimated at roughly **US $8.4 bn (2026) → ~$24 bn (2035), ~8.7 % CAGR** (market-research consensus; figures vary by definition). This is a growing market, not a niche in decline.
- **Illustrative top-down TAM/SAM (our estimate):** ~5 million EU non-residential buildings at €100–700/building/month imply an order-of-magnitude **TAM ~€18 bn/yr**, with a **DACH-focused SAM ~€4.3 bn/yr**. These are illustrative framing figures, not bottom-up forecasts.
- **Serviceable obtainable (3-year, realistic):** 100–200 customers → **€1–2 m ARR**.

5.2 Customers & segments

Bottom-up, validation-first: start where deals close in **2–8 weeks** — university campuses and DACH **mid-market property managers** (5–50 buildings) — then use case studies to reach hotels, healthcare and REITs. Enterprise procurement is deliberately deferred until the product and references are mature. First target: the **BSBI campus pilot**, then one Berlin property manager.

5.3 Business model & pricing (per building / month)

Tier	Price	Target segment
Insight	€99	SME offices
Monitor	€299	mid-market property management
Copilot	€699+	enterprise, healthcare, IoT-equipped
Portfolio (custom)	€5–50k/mo	10+ building portfolios

A pilot is free for the first three months to win reference cases.

5.4 Unit economics (targets, pre-revenue)

These are **targets to validate in the first pilots**, not results:

- ARR per customer ≈ **€17k** (5 buildings × €299/mo).
- Gross margin **~85 %** (SaaS-typical, minus cloud/inference cost).
- Target **CAC payback < 6 months** in a founder-led, content- and network-driven motion.
- Designed to **break even around the second customer** on infrastructure cost.

We deliberately avoid headline LTV/CAC multiples until real acquisition data exists; the figures above are conservative planning assumptions.

5.5 Competitive landscape

Competitor	Position	Gap EnergyLens fills
Measurabl / Aquicore	US enterprise, well-funded	mid-market EU coverage
Cortexa	EU enterprise	mid-market accessibility & price
Schneider EcoStruxure	peer (auditable reporting + AI)	multi-protocol IoT + Fabric-native + mid-market price
Siemens Desigo	BMS / building automation	the analytics/intelligence layer, not a BMS

Our one-line position: *the Microsoft Fabric-native energy-intelligence platform for the EU mid-market, with an AI copilot that reasons over the Lakehouse via tool use — from under €299/building/month.*

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6. 12-month implementation plan

Framing. The technical foundation already exists and the founder builds quickly; the genuine 12-month work is **customer validation, sales, incorporation, regulatory/IP and research** — the things a grant is meant to fund. Milestones are therefore expressed as **validation and business outcomes**, with technical development as the supporting work. Because execution is fast, the plan is front-loaded and milestones can be reached early.

Quarter 1 (months 1–3) — Validate

- **Outcome:** BSBI campus pilot live; 2–3 design partners engaged; pricing hypotheses in test.
- **Build:** production hardening; predictive-maintenance models in training; the carbon-methodology research started with BSBI.
- **Metric:** ≥ 1 live pilot, ≥ 5 discovery conversations/week.

Quarter 2 (months 4–6) — Sell

- **Outcome:** convert pilots into **2–3 paying customers**; pricing validated.
- **Build:** ship the **agentic copilot** (propose-and-sequence) as the commercial wedge; harden the compliance hub for real client data.
- **Metric:** first recurring revenue; ≥ 1 named case study agreed.

Quarter 3 (months 7–9) — Incorporate & build the team

- **Outcome: UG (haftungsbeschränkt) incorporated**; first hire (GTM/sales-engineering); pipeline toward 5 customers.
- **Build:** closed-loop optimisation (battery + HVAC + demand response) in production; security/GDPR review.
- **Metric:** company registered; 4–5 customers signed or in late-stage.

Quarter 4 (months 10–12) — Fundable

- **Outcome: 5+ customers, €5–10k MRR**; auditable-carbon methodology paper submitted with BSBI; Microsoft Partner Network track entered.
- **Build:** scale-readiness (onboarding automation, reliability); seed/angel materials prepared.
- **Metric:** sustainable MRR; a defensible, demonstrable innovation; a credible seed story.

Parallel, before the grant even starts. Because EXIST decisions take several months (apply Jul–Aug 2026 → decision Sep 2026–Feb 2027 → start Q1 2027), the founder will not wait: the BSBI pilot, early outreach and freelance income proceed in the interim, so the venture enters the grant period with **traction already in hand** — which also strengthens the application itself.

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7. Use of funds (≈ €65,000 over 12 months)

EXIST provides a monthly stipend (living costs), a material budget and a coaching budget. Indicative allocation:

Category	Amount	Purpose
Stipend (living costs)	€30,000	12 months full-time focus (rent, living, insurance)
Software & cloud (material)	€5,000	Microsoft Fabric capacity (F-SKU), Power BI Premium, Azure OpenAI inference, SaaS tooling
Hardware (material)	€3,000	Development laptop refresh, monitor, A/V for demos & content
Customer acquisition (material)	€5,000	LinkedIn Sales Navigator, targeted ads, conferences, content production
Legal & administrative (material)	€4,000	UG incorporation, IP/patent counsel, tax advice, GDPR review
Brand & web (material)	€3,000	Brand refinement, marketing website, pitch materials

Category	Amount	Purpose
Pilot deployment (material)	€5,000	On-site setup and sensor integration for BSBI + early customers
Contingency (material)	€5,000	Unforeseen costs
Coaching	€5,000	Startup mentor, sales coaching, technical advisory
Total	**≈ €65,000**	12 months full-time toward an incorporated, revenue-generating company

The **material budget (~€30,000)** is the discretionary portion above; the **stipend (€30,000)** funds the founder's full-time commitment; **coaching (€5,000)** funds external mentoring. Exact figures follow the official EXIST stipend rate for an M.Sc. graduate and the Projektträger Jülich guidelines.

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8. Team, risks & mitigations

8.1 Team & hiring plan

The venture is currently a **solo founder**. EXIST permits founding teams of up to three, and a single founder is a recognised application risk. Mitigation and plan:

- **Founder coverage:** the founder already spans the two hardest-to-combine skill sets (energy domain + data engineering), reducing early dependency on hires.
- **Ecosystem around the founder:** an academic mentor (BSBI), the BSBI Transfer Office, and 1–2 advisors form a support structure that an evaluation panel can see.
- **First hire (Q3):** a GTM / sales-engineering profile to complement the founder's technical strength — explicitly budgeted and timed to first revenue.
- **Co-founder:** actively evaluated; the grant year is the right window to find a complementary commercial co-founder once early traction de-risks equity discussions.

8.2 Risks & mitigations

Risk	Mitigation
Solo-founder bandwidth	First GTM hire in Q3; advisor network; automation of onboarding/support
Pilot → paid conversion is slow	Free, low-risk pilots; quantified value reports; side freelance income as runway buffer

Risk	Mitigation
"Integration, not innovation" perception	The four-axis innovation agenda (agentic AI, predictive, optimisation, methodology IP) + an academic paper provide defensible novelty
Methodology / compliance over-claim	Honest labelling ("ESRS-E1-aligned reporting support", screening estimates flagged); claims reviewed before any "compliant" wording
Visa / residency	18-month post-graduation Job-Search Visa; intent to incorporate (UG) and remain in Germany
Single-vendor (Microsoft) dependency	Provider-abstracted AI layer; standards-based IoT; data portable via open Delta/Parquet
Incumbent price pressure	EU-regulatory depth + Fabric-native lineage + mid-market focus as differentiation, not price

8.3 Honesty layer (current limitations)

We disclose known limitations openly, because a credible application is precise about maturity: Scope 3 is currently a labelled screening estimate (not disclosure-grade); market-based Scope 2 needs supplier-contract data to differ from location-based; CRREM pathways are indicative pending a License-Partner agreement; some HVAC splits are modelled (HDD/CDD method) rather than sub-metered; battery-ROI savings are currently upper-bound estimates pending hour-by-hour tariff matching; building consumption data is synthetic until the first real pilot. Each has a concrete improvement on the roadmap.

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9. Founder & academic anchor

Ali Mert Özdemir.

- **Education:** B.Sc. Energy Engineering (Yaşar University, İzmir); **M.Sc. Energy Management (BSBI Berlin, June 2026)**.
- **Certification: Microsoft DP-600** (Implementing Analytics Solutions Using Microsoft Fabric) — a recent, in-demand certification rarely held by solo founders.
- **Demonstrated execution:** designed and built the entire EnergyLens stack — Fabric medallion pipeline, 9-page Power BI report, IoT/fault-detection layer, battery simulator, web application and AI copilot — **solo, end-to-end, in ~5 weeks**, accelerated by AI-assisted development.
- **Residency:** Berlin-based; 18-month Job-Search Visa runway post-graduation to incorporate and reach first revenue in Germany.

Academic anchor (BSBI). EnergyLens grew directly out of the founder's M.Sc. work at BSBI. The academic mentor provides (a) the **mentor letter required by EXIST**, (b) guidance on the energy-management and methodological dimensions, and (c) a route to the **BSBI campus pilot** and a **joint methodology publication** — giving the venture a genuine research-transfer character. This university anchoring is central both to EXIST eligibility and to the innovation agenda (§4.2, axis M).

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10. Application timeline & next steps

When	Step
Jun–Jul 2026	Mentor letter + BSBI Transfer Office engagement; finalise idea sketch & plan
Jul–Aug 2026	Submit via BSBI to Projektträger Jülich
Sep 2026 – Feb 2027	Evaluation (3–6 months)
Q1 2027	Start of the 12-month grant (target)
In parallel (now → start)	BSBI pilot, early customer outreach, freelance runway — traction before the grant begins

Immediate next steps: (1) confirm the academic mentor and obtain the mentor letter; (2) identify the EXIST coordinator at BSBI (Transfer Office / partner university); (3) start the BSBI campus pilot; (4) finalise this package with the Transfer Office.

EnergyLens — turning every building into a self-diagnosing, decarbonising asset. Draft application package; figures marked target/illustrative to be validated. energylens.eu · Microsoft DP-600.